ANN Tuner

HYPERPARAMETER CONTROL

Architecture

Hidden Layers

3

Layer 1 Neurons

128

Layer 2 Neurons

64

Layer 3 Neurons

32

Training

Activation

UPI Fraud Detection

Neural Network Dashboard

◈ ANN · HYPERPARAMETER TUNING · LIVE TRAINING · SEVERE IMBALANCE (0.15%) · SMOTE · CLASS WEIGHTS

⚠️ **SEVERE CLASS IMBALANCE DETECTED** — Fraud rate: **0.152%** (76 fraud out of 50,000 transactions)

📌 Standard accuracy is **misleading** here — focus on **ROC-AUC, Recall & F1-Fraud**.

📌 SMOTE + Class Weights + **lowered decision threshold (0.3)** are enabled to handle this.

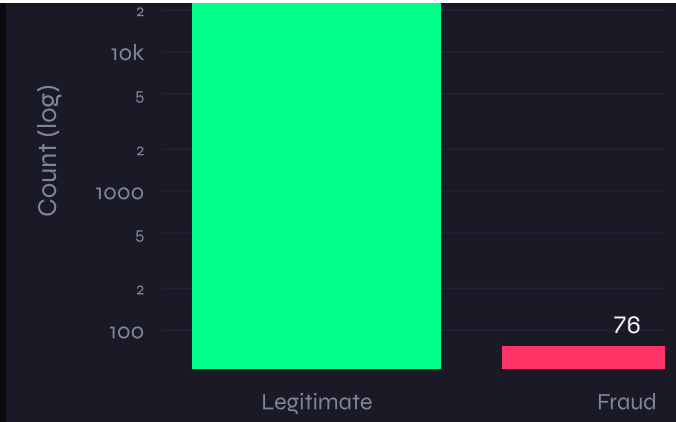
📊 EDA & OVERVIEW | 🧠 TRAIN ANN | 📈 RESULTS & METRICS | 🔍 LIVE PREDICTOR

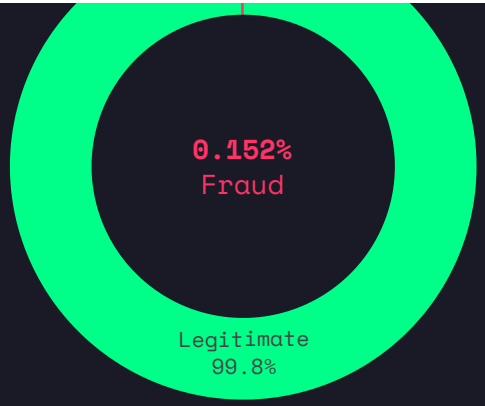
50,0	76	49,9	0.15	₹1,3	₹1,7
TOTAL TXNS	FRAUD CASES	LEGIT CASES	FRAUD RATE	AVG AMOUNT	AVG FRAUD AMT

CLASS IMBALANCE — THE CORE CHALLENGE

Class Distribution (Log Scale) — 76 fraud vs 49,924 legit

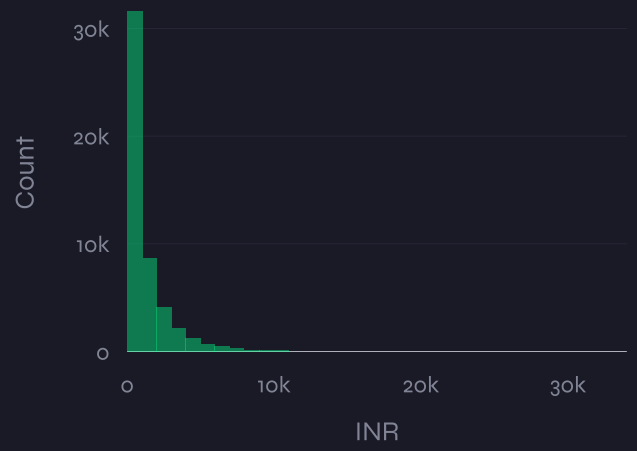
Extreme Imbalance — Why SMOTE & Threshold Tuning Matter



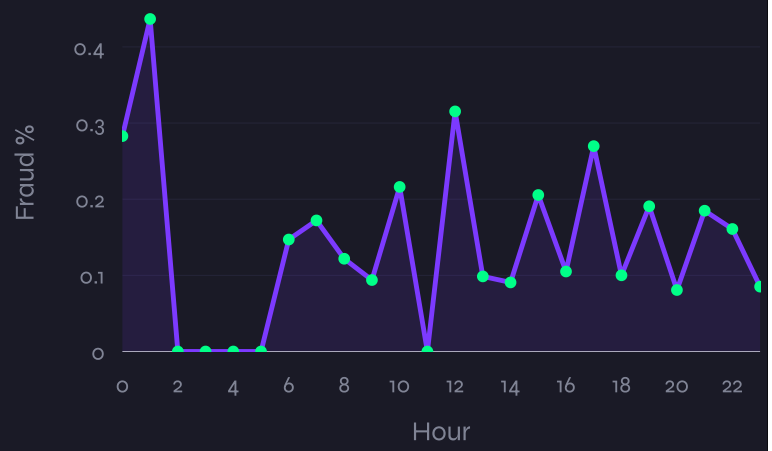


FRAUD PATTERN ANALYSIS

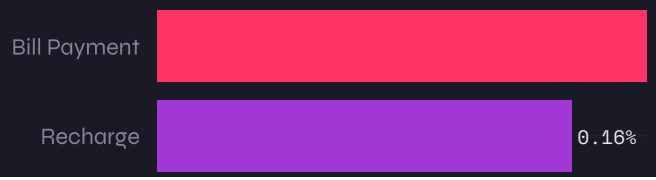
Amount Distribution by Class



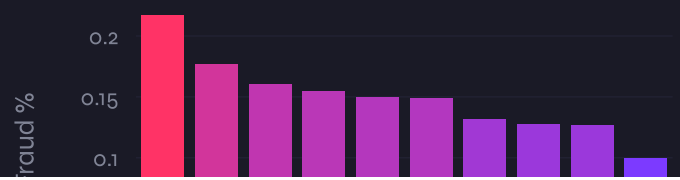
Fraud Rate by Hour of Day

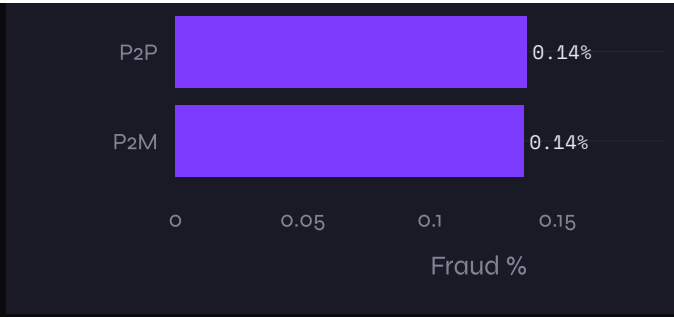


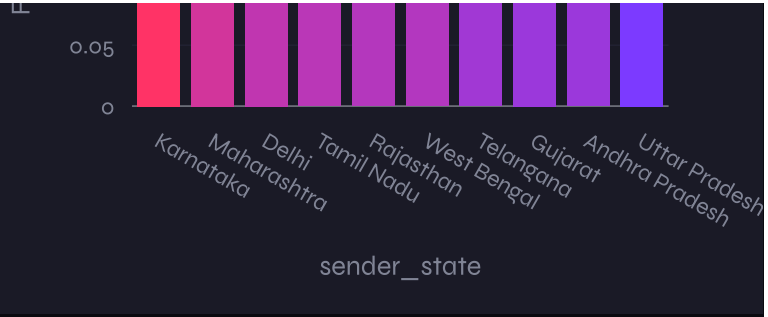
Fraud % by Txn Type



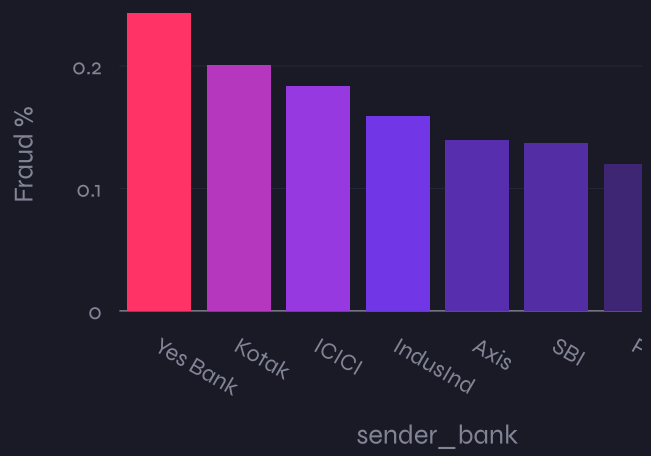
Top 10 States — Fraud Rate



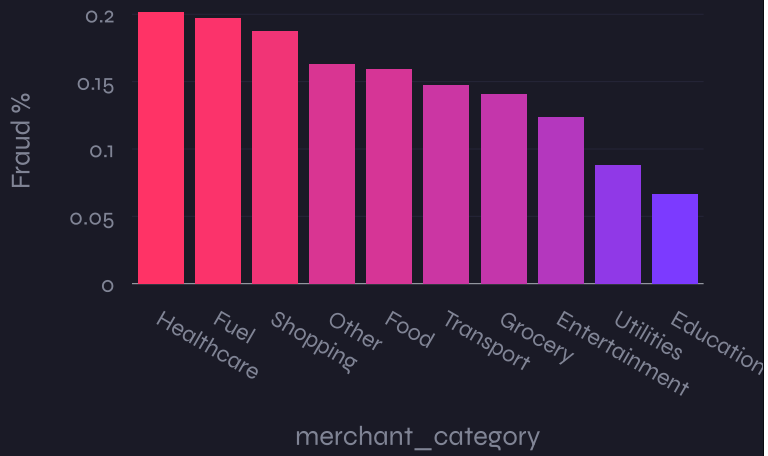




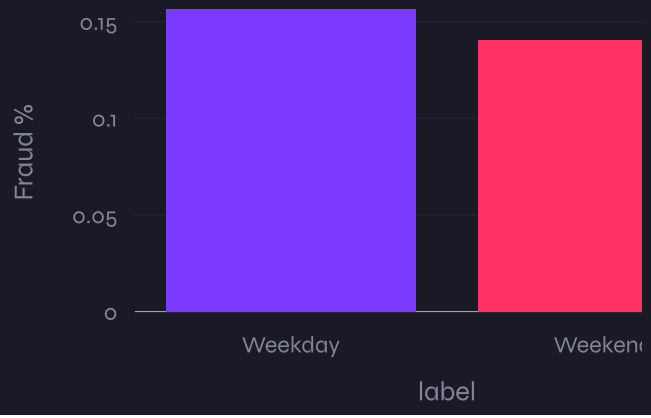
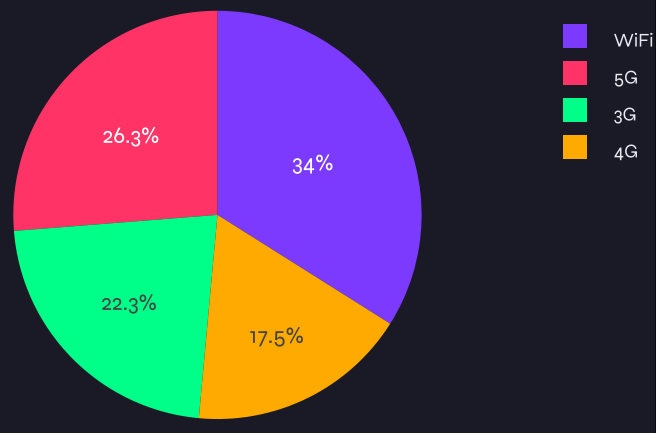
Fraud % by Bank



Fraud % by Merchant Category



Fraud Rate: Weekday vs Weekend



Fraud Share by Network Type